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Palaeomagnetic results from Early Mesozoic strata in the Qaidam Basin and their implications for the formation of the Northern China Domain

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SUMMARY

The Northern China Domain is located between the Central Asian Orogenic Belts to the north and the Kunlun–Qinling belt to the south, and it comprises the North China, Alxa and Tarim blocks. The relationships among the Northern China domain and the southern tectonic elements such as the Qaidam Basin/Terrane are debated because of the major modification by crustal deformation in the late Mesozoic–Cenozoic. To address this issue, we conducted a palaeomagnetic and high-precision radiometric dating study of Triassic volcanic rocks and Middle Jurassic strata in the Qaidam Terrane. Our objective was to determine the relationship between the Qaidam Terrane with the Tarim Block and the North China Block (NCB) during the late Palaeozoic and early Mesozoic. Four volcanic samples yielded zircon U-Pb ages of 236–243 Ma. The characteristic remanent magnetizations (Middle Triassic: $D = 40.2°$, $I = 54.6°$, $\alpha_{95} = 3.4°$; Middle Jurassic: $D = 27.4°$, $I = 48.0°$, $\alpha_{95} = 7.9°$) passed the fold and reversal tests, and yielded Middle Triassic and Middle Jurassic palaeopole positions at 57.6° N, 178.2° E, $A_{95} = 4.0°$ and 65.8° N, 197.6° E, $A_{95} = 7.8°$, respectively. Based on these new poles, combined with other reliable data, we compared the apparent polar wander path (APWP) of the Qaidam Terrane with those of the NCB and Tarim Block. The results show that, from the Carboniferous through Early Cretaceous, the APWP of the Qaidam Terrane resembles that of the Tarim Block, but it is quite different from that of the NCB. Combined with other reported evidence, we conclude that the Qaidam Terrane was an independent dynamic unit during the late Palaeozoic until its connection with the Tarim Block, which was followed by continuous eastward motion. During this process, the connection between the Qaidam Terrane and the NCB–Alxa blocks occurred in the Middle Triassic, and subsequently the Qaidam Terrane underwent multiple tectonic responses to collisions with the Qiangtang Terrane, Lhasa Terrane and the India Plate, before the formation of its modern tectonic configuration.

Key words: Asia; Plate motions; Palaeomagnetism.

1. INTRODUCTION

Eastern Asia consists of a mosaic of terranes and blocks that were distributed across the eastern Palaeo-Tethys Ocean during the Late Palaeozoic. From the Permian onwards, these terranes/blocks began to converge and were finally amalgamated during the Triassic. This process left geological traces—including multiple ophiolitic oceanic crust remnants, orogens and sutures—implying a complex tectonic evolution that included dispersal, drift and the suturing of continental fragments and the intervening oceanic crust (Metcalf 1994, 1996, 2013; Xu *et al.* 2013; Zhang *et al.* 2013a; Huang *et al.* 2018; Zhao *et al.* 2018). Large-scale tectonic deformation within the Altyn–Tagh and Qilian Shan fold belts occurred during the late Cenozoic (Yin & Harrison, 2000), related to the India–

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Eurasia collision and crustal shortening within the Central Asian Orogenic Belts (CAOB) during the Mesozoic. Hence, the process of reconstructing the pre-Mesozoic tectonic framework has been challenging, and the relationships among terranes/blocks in northern China, including the Qaidam, Tarim, Alxa and North China blocks (NCBs), remain unclear. For example, the similarity of early Proterozoic metamorphic rocks and the late Proterozoic–Palaeozoic deformed sedimentary basement, together with the stratigraphic and sedimentary evolution during late Sinian to early Palaeogene time, has caused several workers to propose that the Qaidam Terrane originally formed part of the Tarim Block and reached its present-day position in the course of strike-slip along the Altyn fault system during the latest Cretaceous through early Cenozoic (Meng *et al.* 2001; Metcalfe 2006). However, the early Palaeozoic high-pressure/ultrahigh-pressure metamorphic rocks and ophiolitic melange in the Altyn Tagh system (Liu *et al.* 1999, 2018; Li *et al.* 2009), which presently separates the Tarim and Qaidam blocks, imply that these two tectonic units were separated by the Proto-Tethys Ocean prior to their amalgamation in the early Palaeozoic (Zhao *et al.* 2018). Furthermore, the clearly different palaeolatitudes imply their renewed separation during the Early Carboniferous (Wang *et al.* 2016). Moreover, there is still considerable debate about the relationship among the three blocks of the Northern China Domain: the Qaidam, Alxa and NCBs. Several authors have suggested that the Qaidam Terrane had already accreted to the Qilian Orogen, Alxa/Hexi corridor Terrane and the NCB by the late Devonian, following the closure of the Qilian Ocean, an eastern branch of the Proto-Tethys Ocean (Xiao *et al.* 2009; Cocks & Torsvik 2013; Song *et al.* 2013; Zhao *et al.* 2018). However, Yuan & Yang (2015a,b) suggested that the Alxa Terrane, which is located in between the Qaidam terrane and the NCB, was not part of the NCB during the late Devonian and that it migrated to the NCB via 32° counterclockwise rotation after the Early–Middle Triassic along a tectonic boundary located between the Helanshan and Zhuozishan Mountains. The Permian–Triassic interval was the key to the formation of the East Palaeo–Tethys Orogenic Belt and CAOB. Therefore, it is necessary to understand the relationship between the Qaidam and Tarim blocks, as well as the relationship between the Qaidam, Alxa and NCBs during this interval, because of important geological implications for understanding the evolutionary processes of these associated orogenic belts. Moreover, Li *et al.* (2018) suggested that the Qaidam and Tarim blocks were originally a single tectonic unit which later underwent large-scale orogeny that transformed the original linearly distributed Proto-Tethys subduction zone to current circular distribution around the Qaidam Terrane. Therefore, understanding the tectonic affiliation between the Qaidam Terrane and the surrounding blocks and terranes has significant implications for understanding the convergence process of Palaeozoic and Mesozoic plates.

Palaeomagnetic techniques have been widely used to assess the drifting and amalgamation histories of terranes. A comparison of the palaeolatitudes or apparent polar wander paths (APWPs) of two terranes can reveal whether they belonged to a unified dynamic unit. Although the APWPs of the Tarim Block and NCB (Zhu *et al.* 1998; Yang et al., 1998; Yang & Besse 2001; Huang *et al.* 2008; Wu *et al.* 2017) have already been constructed in detail, palaeopoles from the Qaidam Terrane are limited, especially for the Permian and Triassic, which are important periods for evaluating the convergence of related blocks and terranes, as well as for determining the tectonic implications for northern China. In this context, we conducted geochronological and palaeomagnetic studies of Triassic volcanic rocks and Jurassic sedimentary strata from the Qaidam Terrane, with the objective of supplementing the available APWPs and then to assess the relationships among the relevant blocks during the late Palaeozoic through early Mesozoic.

2. GEOLOGICAL SETTING

The Qaidam Terrane is separated by the Altyn fault system to the northwest, the Qilian Orogen to the northeast and the Eastern Kunlun Mountains to the south. It consists of metamorphic Proterozoic crystalline basement, and a low-grade metamorphic or unmetamorphosed sedimentary cover that includes Cambrian and Ordovician shallow marine facies, Late Devonian continental deposits and Carboniferous marine sediments, while Silurian, Permian and Triassic strata are generally lacking. Mesozoic strata include Jurassic, Cretaceous and Cenozoic continental clastic rocks (BGMRQP 1991). Fortunately, Triassic volcanic rocks are exposed at the eastern and southern edges of the block. The western exposure is in the Qimantagh region, and it comprises mainly Upper Triassic continental volcanic rocks, including brick red, dark purple or grey rhyolites interbedded with dacite that unconformably overlie the underlying strata. The eastern exposure is in the Dulan area and comprises mainly andesites and interbedded dacite tuff and tuff breccias (Fig. 1b, BGMRQP 1991). These volcanic rocks occur along the Permian–Triassic Eastern Kunlun–Qaidam magmatic belt and are considered to have originated from subduction and the collision of the Palaeo-Tethyan Ocean basin (Chen *et al.* 2010, 2015). The Jurassic sedimentary rocks are entirely nonmarine and are mainly distributed in the northern and northeastern parts of the Qaidam Basin (BGMRQP 1991). The Jurassic strata are assigned to four formations (Fms): the Xiaomeigou, Dameigou, Caishiling and Hongshuigou Fms, from bottom to upper. The Lower Jurassic (Xiaomeigou Fm) is 100 m thick and is overlain with an angular unconformity by Middle Jurassic rocks(Dameigou Fm and Caishiling Fm). The Upper Jurassic (Hongshuigou Fm) conformably overlies the Middle Jurassic (BGMRQP 1991). Our palaeomagnetic samples were collected from the Caishiling Fm, which is composed mainly of purple sandstone and brick red mudstone, and the contents of sporopollenin (*Classopollis*) and Ostracods (*Darwinula sarytirmenensis, Timiliasevia catenullaria*) indicate a Middle Jurassic age (BGMRQP 1991).

3. SAMPLING AND METHODS

Zircon grains from four igneous bulk samples were extracted for U-Pb dating. Samples DLT01 and DLTB02 are from Dulan, and samples WD01 and WD02 are from the Wutumeiren region (the names are the same as those of the palaeomagnetic sampling sites). The zircons from three samples (WD01, WD02 and DLT01) were analysed using a sensitive high-resolution ion microprobe (SHRIMP) at ANU. Uncertainties in the calculated weighted mean ages for the pooled data are reported with 95 per cent confidence limits which include the uncertainties in the standard calibrations where appropriate (see details of the standard calibrations for individual samples in the data tables); they and were calculated using the SQUID 2 Excel macro of Ludwig (2009). One sample (DLTB02) was analysed at Northwest University, China, using an Agilent 7500a ICP-MS equipped with a 193 nm ArF-excimer laser (MicroLas Pro). A 32 μm spot diameter was used in this study, with a laser repetition rate of 6 Hz. Zircon 91 500 (Wiedenbeck *et al.* 1995), analysed after every five analyses of the zircon grains, was used as an external standard to correct for both instrumental

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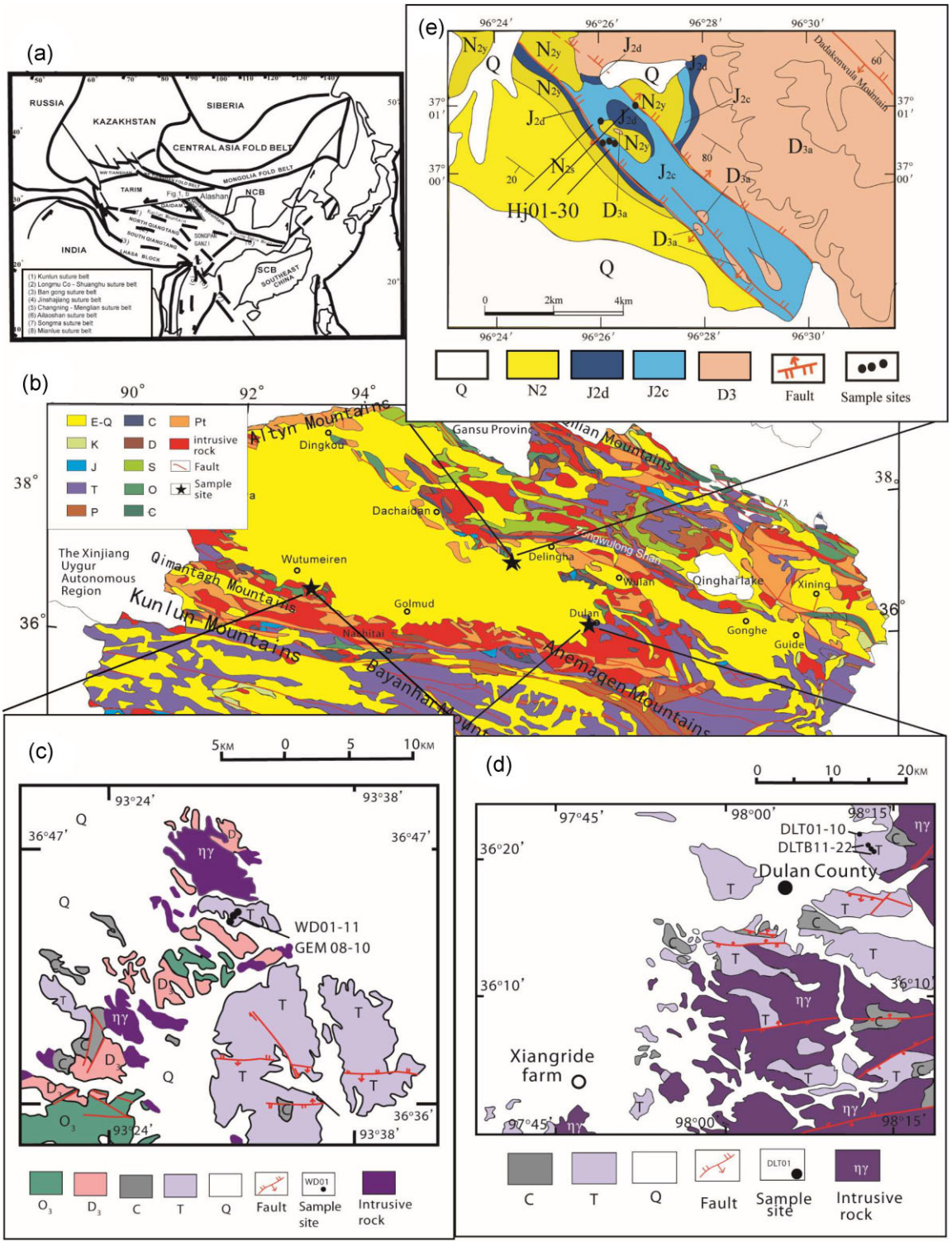


Figure 1. (a) Simplified tectonic map of China and the adjacent regions (modified from Yang & Besse 2001). The star indicates the sampling location. (b) Structural outline of the Qaidam Basin and its neighbouring areas (modified from Ma *et al.* 2002). (c)–(e) Simplified geological map of the sampled location and geological cross-sections in the Wutumeiren, Dulan and Olongbuluke regions, respectively (modified from unpublished 1:200 000 geological maps). Abbreviations: Q, Quaternary; E, Palaeogene; K, Cretaceous; J, Jurassic; T, Triassic; P, Permian; C, Carboniferous; D, Devonian; S, Silurian; O, Ordovician; Є, Cambrian and Pt, Proterozoic.

mass bias and depth-dependent elemental and isotopic fractionation. NIST SRM610 was analysed twice every 10 analyses for U, Th and Pb concentration measurements. Raw data were processed using the GLITTER 4.0 program (van Achterbergh *et al.* 2001). U-Pb ages were calculated using IsoplotEx 3 software (Ludwig 2003).

The zircon grains extracted from the basaltic andesite sample were euhedral. The cathodoluminescence (CL) images of

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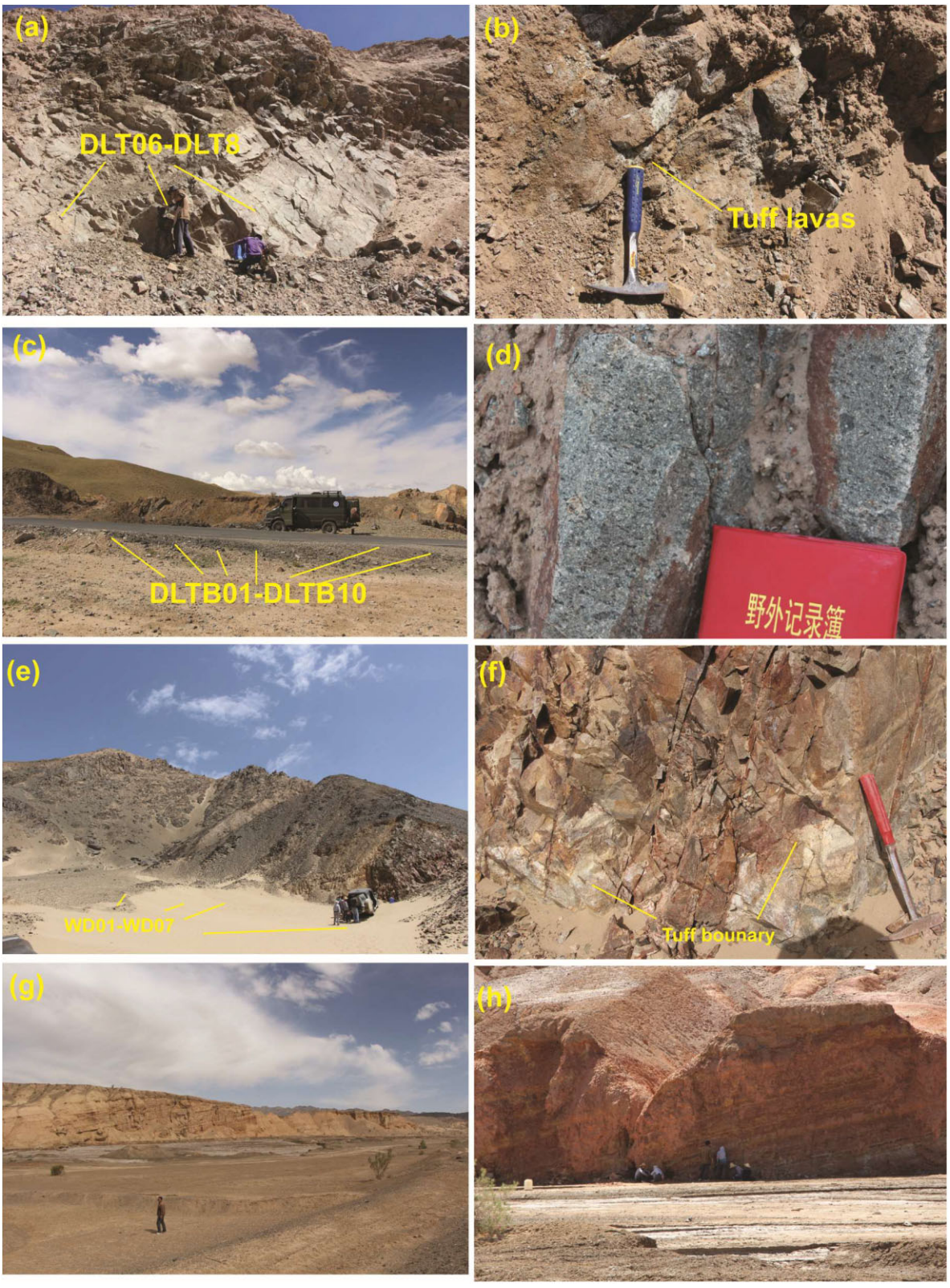


Figure 2. (a), (c), (e) and (g) Representative photos of the sampled sections, (b) and (f) tuff and (d) flotation structure.

the analysed zircons show oscillating zonation (Figs 3a1–d1), and the Th/U ratios of almost all zircons are > 0.3 (Tables A1–A4, Supporting Information). These characteristics indicate that the zircons are of igneous origin. The U-Pb age data are presented in Fig. 3 and Tables A1–A4 in Supporting⟨?PMU ?⟩ Information.

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Totals of 139 samples (14 sites), 181 samples (22 sites) and 309 samples (30 sites) were collected from andesite lavas and sedimentary strata near Wutumeiren Village (36.7° N, 93.5° E), Dulan County (36.4° N, 98.2° E) and Olongbuluke Village (Figs 1c–e), respectively. Well-developed elongated dark inclusions and tuff deposits indicated the orientation of lava flows (Fig. 2). A gasoline-powered drill was used to obtain the palaeomagnetic samples, which were orientated with magnetic and solar compasses. The present-day geomagnetic declination in the sampled areas (359.8° at Wutumeiren, 0.2° at Dulan and 0.2° at Olongbuluke) was calculated using the International Geomagnetic Reference Field (IAGA 2010). The samples were measured in the palaeomagnetic laboratory of the Australian National University (ANU), Canberra, Australia; at the Institute of Earth Environment, Chinese Academy of Sciences, Xi'an, China; and at Nanjing University, Nanjing, China. All the measuring equipment was installed within a magnetically shielded room with a residual field of ~300 nT, and the residual field inside the thermal demagnetization oven was < 50 nT. To identify the dominant remanence carriers, isothermal remanent magnetization (IRM) was imparted along three orthogonal axes and then subjected to thermal demagnetization, using the procedure described by Lowrie (1990). Composite IRMs were imparted in fields of 0.12, 0.4 and 2.2 T along the *x*-, *y*- and *z*-axes, respectively. Stepwise thermal and alternating-field (AF) demagnetization of the natural remanent magnetization (NRM) was carried out using the following steps: thermal: 100, 200, 250, 300, 350, 400, 450, 480, 500, 520, 540, 555, 570, 580, 590 and 600° C; AF: 2, 3, 4, 6, 8, 10, 12, 15, 18, 21, 25, 30, 35, 40, 45, 50, 60, 70, 80 and 99 mT. The NRM of the volcanic samples were measured using an AGICO JR6 spinner magnetometer, while sedimentary samples were measured using a 2G-755 cryogenic magnetometer. Characteristic remanent magnetization (ChRM) directions and site-mean directions were determined using principal component analysis (Kirschvink 1980) and Fisher (1953) statistics. Different magnetic components were determined based on well-defined linear segments including at least three demagnetization steps on the orthogonal vector diagram, with the maximum angular deviation (MAD) < 15°. Any outliers of the isolated ChRM directions that deviated far from the site average were excluded from further calculations.

4. ZIRCON U-PB DATING RESULTS

For sample DLT01, 22 analyses were carried out using SHRIMP on 21 zircons, except for grain 19 which was measured at the core and rim because of its core-rim structure (Fig. 3, a1). The ^{206} Pb/^{238} U age of the inherited core (19.2) is 413 ± 10 Ma, whereas the rim (19.1) age is 234 ± 3 Ma, which is consistent with the other zircon ages. The 20 analyses (except for 4.1, which was rejected) yielded a weighted-mean ^{206} Pb/^{238} U age of 236.5 ± 1.9 Ma (2σ).

For sample DLTB02, 36 zircon grains were analysed using LA-ICP-MS and only 24 zircons had concordant ages (concordia degree between 110 and 90 per cent). The analyses of the 24 zircons yielded a weighted-mean ^{206} Pb/^{238} U age of 239.0 ± 1.0 Ma (2σ).

For sample WD01, 26 analyses were carried out using SHRIMP on 21 zircons. The 21 analyses (3.1, 4.1, 8.1, 7.1 and 18.1 were rejected) yielded a weighted-mean ^{206} Pb/^{238} U age of 243.3 ± 1.4 Ma (2σ).

For sample WD02, 20 analyses were carried out using SHRIMP on 20 zircons. The 20 analyses yielded a weighted-mean ^{206} Pb/^{238} U age of 240.6 ± 1.3 Ma (2σ).

The ^{206} Pb/^{238} U ages of zircons from these volcanic rocks ranged from 236 to 243 Ma (Middle Triassic ages for the Wutumeiren samples, and Middle to very early Late Triassic ages for the Dulan samples).

5. PALAEOMAGNETIC RESULTS

IRM acquisition was rapid below 200 mT and reached saturation above 800 mT (Figs 4a and b). The thermal demagnetization of three-component IRMs indicated two major unblocking temperatures, 350 and 580° C, for the samples from the Wutumeiren region (Fig. 4d), and 580 °C for the samples from the Dulan region (Fig. 4e). These results indicate that titanomagnetite and magnetite are the dominant remanence carriers in the Wutumeiren samples, whereas magnetite is the dominant carrier in the samples from the Dulan region. For the Jurassic samples, the IRM acquisition curve did not reach saturation, even at 2.2 T (Fig. 4c). Subsequent thermal demagnetization yielded the high unblocking temperature of 680–690 °C (Fig. 4f), indicating that haematite is the dominant carrier.

5.1 Triassic results

After removing the viscous remanent magnetization (VRM), a stable component was isolated from 28 sites (166 samples), within the temperature range of 580–590 °C, or at 99 mT (Figs 5a-e). This VRM (before tilt correction: D_g/I_g = 355.7°/50.7°, k = 51.2 and α_{95} = 5.4°; after tilt correction: D_s/I_s = 358.9°/17.6°, k = 27.5 and α_{95} = 7.4°; Figs 6a and b) was isolated at ~160–480 °C, and it did not pass the fold test (McFadden 1990; Enkin 2003). The stable components passed the fold test (McFadden 1990; Enkin 2003, Figs 6c–e). The parameters for the McFadden (1990) fold test were $\xi_{2\ (\text{in situ})}$ = 10.07 in geographic coordinates and $\xi_{2(\text{tilt corrected})}$ = 1.494 after tilt correction, and the critical value was ξ_c = 6.155 at the 95 per cent confidence level. The fold test of Enkin (2003) showed that optimal clustering of the palaeomagnetic directions occurred at 102 per cent ± 13 per cent unfolding. The result of the reversal test was negative (Norm: Dm = 46.2, Im = 54.8, n = 21, k = 126.7; Rev: Dm = 201.5, Im=–49.4, n = 7, k = 115.2; angle between the average gamma = 16.0), and we suspect that these samples were collected from three lava sections with different ages and thus they failed the reversal test. The directions recorded individual geomagnetic spot fields indicating that our samples had precisely recorded the geomagnetic direction during the formation of these volcanic rocks. The distance between the Dulan and Wutumeiren regions is ~450 km and therefore we calculated the virtual geomagnetic pole (VGP) of all 28 sites (57.6° N, 178.2° E, A_{95} = 4.0°) was calculated (Table 1). The mean direction of 28 sites (D = 40.2°, I = 54.6°, a_{95} = 3.4°) was calculated based on the location of Dulan County (36.4° N, 98.2° E).

Palaeomagnetic results from volcanic samples are usually spot records of the geomagnetic field, and thus it is necessary to determine whether the palaeosecular variation (PSV) has been averaged out. We checked the reliability of our results using two methods to assess their stability (McFadden *et al.* 1991; Biggin *et al.* 2008; Deenen et al 2011): (1) in the method of Deenen *et al.* (2011), the results for the two sampling areas were as follows: A_{95} = 1.9 (124 samples), $A_{95\text{max}}$ = 3.915259 and $A_{95\text{min}}$ = 1.739471 for the Dulan area, and A_{95} = 3.1 (40 samples), $A_{95\text{max}}$ = 7.902418 and $A_{95\text{min}}$ = 2.716863 for the Wutumeiren area. These results suggests that the PSV of our data from volcanics has been averaged out. (2) In the method

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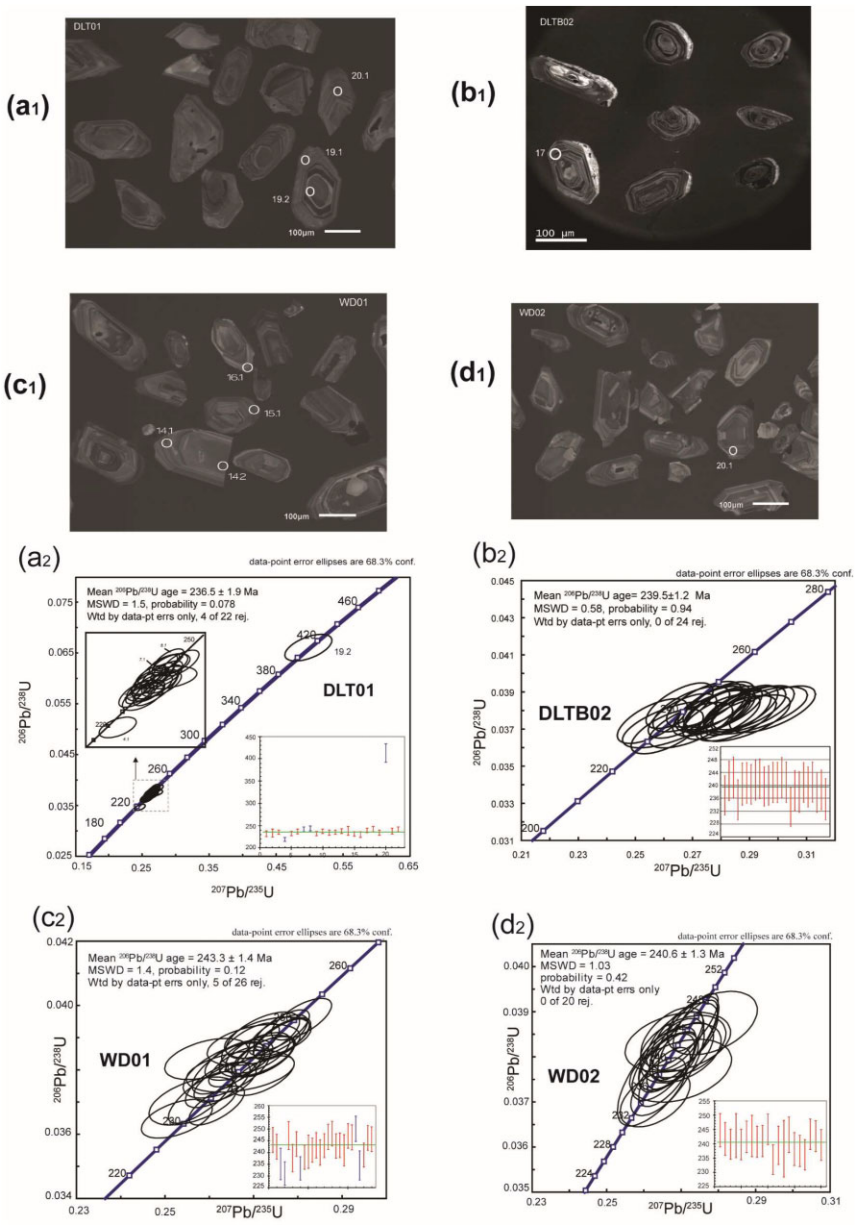


Figure 3. CL images of zircons from samples (a1) DLT01, (b1) DLTB02, (c1) WD01and (d1) WD02, and Concordia diagrams of zircon analyses for samples (a2) DLT01, (b2) DLTB02, (c2) WD01and (d2) and WD02. Circles with numbers are the spot position and their number. The ellipse is an error ellipse. Wtd: weighted, and MSWD: mean squared weighted deviation.

proposed by McFadden *et al.* (1991) and Biggin *et al.* (2008), our calculated S_B value (11.3) is within the expected range (11.3–17.9), which was estimated from the Model G shape parameters during the Triassic (Handford *et al.* 2021) suggesting that our sampling of the studied lavas has averaged out the PSV. Furthermore, the VGP of all 28 sites (57.6° N, 178.2° E, $A_{95} = 4.0°$) for the Middle Triassic rocks is different from the reported younger palaeomagnetic poles from the Qaidam Basin (Neogene: 86.5° N,254.2° E, $A_{95} = 5.7°$; 79.9° N, 274.5° E, $A_{95} = 3.2°$; Eocene: 63.4° N, 211.2° E, $A_{95} = 9.9°$; Early Cretaceous: 75.2° N, 182.3° E, $A_{95} = 7.1°$; Late Jurassic: 61.7° N, 319.7° E, $A_{95} = 5.4°$; 50.1° N, 197.2° E, $A_{95} = 7.3°$, Chen *et al.* 2002; Sun *et al.* 2006). Thus, we consider that the magnetization of the Middle Triassic volcanic rocks at all 28 sampling sites is primary.

5.2 Jurassic results

Among the 309 samples collected, single stable magnetization components were isolated from only 89 samples during thermal demagnetization up to 690 °C (Figs 5f–h). The site mean direction in geographic coordinates is: $D_g/I_g = 25.0°/43.4°$, $k = 18.5$ and $\alpha_{95} = 10.4°$; and in tilt-corrected coordinates, the site mean direction is: $D_s/I_s = 27.4°/48.0°$, $k = 30.8$ and $\alpha_{95} = 7.9°$ (Figs 7a and b, and Table 2). Fold (McFadden 1990; Enkin 2003) and reversal tests (McFadden & McElhinny 1990) were applied to this single component direction. Although the result of the fold test (McFadden 1990) was inconclusive, possibly because of the almost horizontal orientation of sampled strata, the maximum of the precision parameter was achieved at 100 per cent unfolding(Fig. 7c). This direction passed

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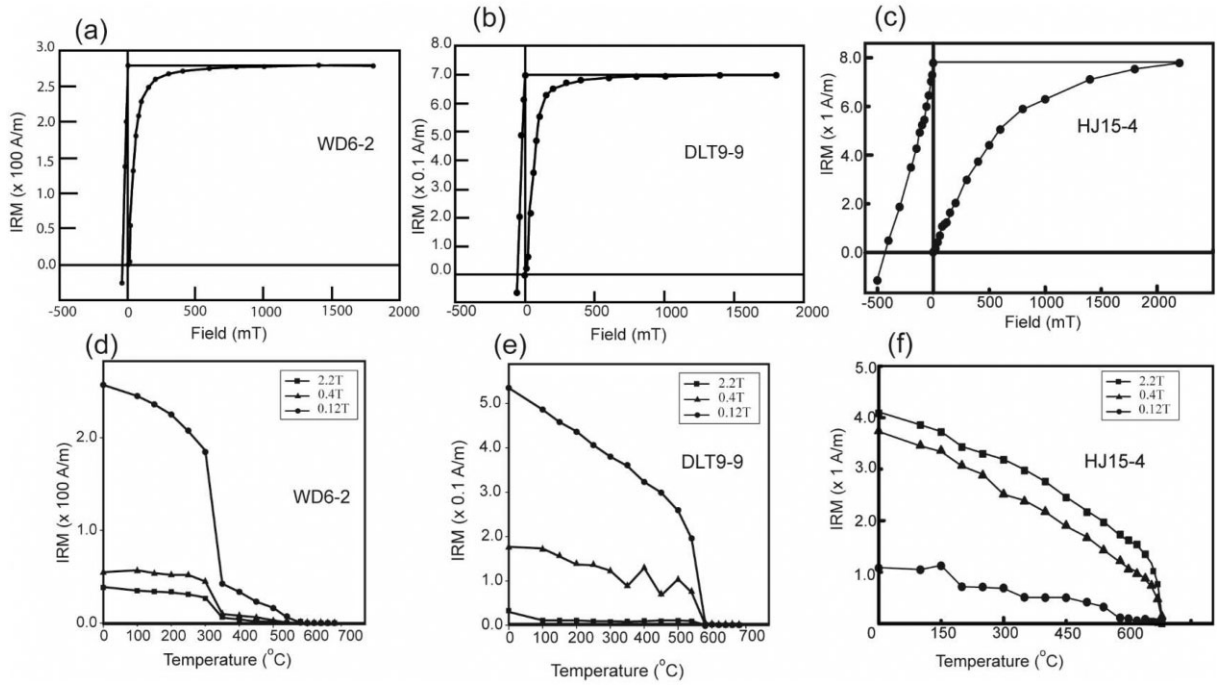


Figure 4. IRM acquisition curves for samples from the Qaidam Basin. (a) and (c) Note the rapid increase below 150 mT and saturation above 800 mT. Three-component IRM thermal demagnetization showed major unblocking at (b) ∼350 and ∼580 °C for a lava sample from the Wutumeiren region and (d) 580° C for a sample from the Dulan region. These results suggest that titanomagnetite and magnetite are the dominant magnetic carriers in the samples from Wutumeiren, while magnetite is the dominant carrier for the Dulan region. (e) For Jurassic samples, the IRM acquisition curve did not reach saturation even at 2.2 T. (f) Subsequent thermal demagnetization yielded the high unblocking temperature of 680–690 °C, implies that the haematite is the domiant carrier of these samples from the Olongbuluke region.

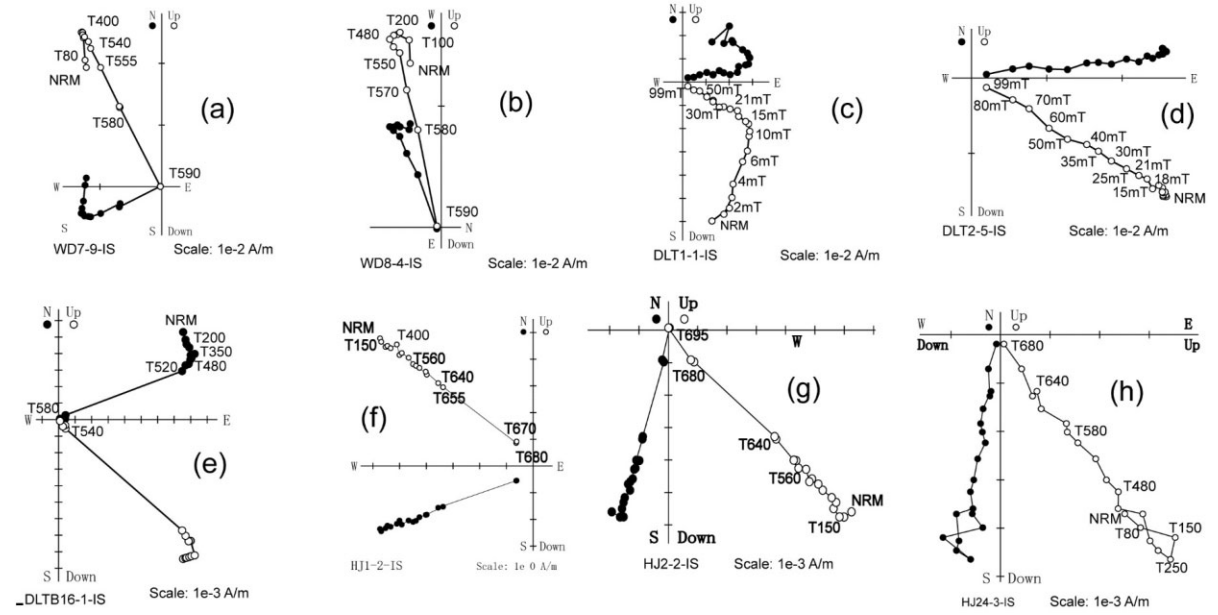


Figure 5. Representative vector demagnetization diagrams for thermal and AF demagnetization of the NRM of the studied Triassic samples (*in-situ* coordinates). The solid and open symbols indicate vector endpoints projected onto the horizontal and vertical planes, respectively.

the fold test (Enkin 2003), and the optimal clustering of palaeomagnetic directions occurred at 116 ± 49 per cent unfolding. The result of the reversal test was positive in class C. The mean direction of the normal polarity is $Dm/Im = 24.5/46.2$, $n = 5$ and $k = 42.1$; the mean direction of the reversed polarity is $Dm/Im = 209.6/-49.2$, $n = 7$, $k = 23.7$; and the angle between the mean directions $\gamma = 4.6° < \gamma_{critic} = 16.8$. The site mean direction of all 12 sites(81 samples) corresponded to a pole (65.8° N, 197.6° E, A95 = 7.8°), which is different from the reported younger palaeomagnetic poles from the Qaidam Basin (Neogene: 86.5° N, 254.2° E, $A_{95} = 5.7°$; 79.9° N, 274.5° E, $A_{95} = 3.2°$; Eocene: 63.4° N, 211.2° E, $A_{95} = 9.9°$; Early Cretaceous: 75.2° N, 182.3° E, $A_{95} = 7.1°$; Late Jurassic: 61.7° N,

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Table 1. ChRM directions for Middle Triassic volcanic rocks in the Qaidam Basin.

Site	Lat. (° N)	Long. (° E)	Strike/dip	*N*/*n*	D_g (°)	I_g (°)	D_s (°)	I_s (°)	*k*	α_{95} (°)	VGP		
											Λ (° N)	Φ (° E)	*Dp*/*dm*
DLTB11	36°21.743′	98°12.282′	198/24	7/8	70.7	39.0	51.3	55.7	124.1	5.4	49.2	172.5	5.5/7.7
DLTB12				7/10	59.3	31.1	43.5	44.6	32.5	10.8	51.8	191.0	8.6/13.6
DLTB13				10/10	62.0	33.3	45.1	47.6	81.9	5.4	51.6	186.4	4.6/7.0
DLTB14				4/9	67.9	40.3	46.7	55.9	44.7	13.9	52.8	173.8	14.3/19.9
DLTB15				9/9	59.6	38.5	38.7	51.3	54.7	7.0	57.9	184.6	6.4/9.5
DLTB16				8/8	61.7	40.0	39.8	53.4	165.6	4.3	57.6	180.5	4.2/6.0
DLTB17				9/13	55.0	36.7	35.2	48.0	67.5	6.3	59.7	192.0	5.4/8.2
DLTB18				10/11	54.1	36.4	34.4	47.3	369.0	2.5	60.1	193.6	2.1/3.2
DLTB19				5/10	68.8	41.6	46.8	57.4	85.5	8.3	53.1	171.2	8.9/12.1
DLTB20				6/11	61.9	37.2	42.3	50.9	87.4	7.2	54.9	183.4	6.6/9.7
DLTB21				5/9	63.8	24.0	52.2	39.6	33.0	13.5	43.1	190.4	9.7/16.2
DLTB22				4/8	72.5	40.4	52.4	57.6	83.3	10.1	48.9	169.2	10.9/14.8
DLT1	36°21.866′	98°12.012′	205/40	6/9	83.2	30.0	53.1	58.9	177.1	5.0	48.7	167.0	5.6/7.5
DLT2				4/5	84.0	29.6	54.5	59.1	509.1	4.1	47.7	166.3	4.6/6.1
DLT4				3/9	81.7	31.2	49.6	58.9	130.9	10.8	51.3	167.8	12.0/16.1
DLT5				4/10	81.3	31.2	49.2	58.7	768.0	3.3	51.5	168.3	3.6/4.9
DLT6				3/8	81.0	34.8	43.8	61.0	802.7	4.4	56.0	164.9	5.2/6.7
DLT7				3/8	85.7	36.2	46.9	64.8	217.8	8.4	54.0	156.6	10.9/13.5
DLT8				4/9	83.3	30.9	51.9	59.6	479.3	4.2	49.7	166.1	4.8/6.3
DLT9				4/10	81.9	31.8	49.1	59.4	1101.0	2.7	51.8	167.1	3.0/4.1
DLT10				9/11	81.8	29.9	51.4	58.0	86.6	5.6	49.7	168.9	6.1/8.3
Mean				21	71.2	35.2	46.0	54.6	122.6	3.0	53.0	176.1	2.9
WD2	36°43.788′	93°31.681′	251/31	7/8	241.8	− 67.3	196.8	− 49.3	86.1	6.5	− 74.6	23.6	5.7/8.6
WD3				7/7	258.5	− 69.5	199.1	− 55.7	221.2	4.1	− 74.7	359.5	4.2/5.9
WD4				4/8	248.2	− 66.7	199.9	− 51.1	197.5	6.6	− 72.9	14.3	6.0/8.9
WD6				4/10	252.0	− 61.2	208.4	− 49.0	119.1	8.5	− 65.4	11.2	7.4/11.2
WD7				7/11	249.2	− 58.6	213.4	− 44.9	63.3	10.5	− 60.0	13.7	8.4/13.3
GEM8				4/8	216.4	− 64.6	188.7	− 40.4	120.0	8.4	− 74.4	62.3	6.1/10.1
GEM10				7/12	257.0	− 65.2	205.1	− 53.2	119.4	5.5	− 69.3	4.2	5.3/7.6
Mean				7	246.2	− 65.3	201.5	− 49.4	115.2	5.6	71.0	197.0	5.5
Mean all sites				**28**			**40.2**	**54.6**	**65.3**	**3.4**	**57.6**	**178.2**	**4.0**

Notes. Lat/Long: latitude and longitude of the sampling site; *N*/*n*: number of samples used to calculate the mean/measured palaeomagnetic direction; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigraphic coordinates, respectively; *k*: precision parameter and α_{95}: half angle of the cone of 95 per cent confidence about the mean direction after tilt correction. The parameters for the McFadden (1990) fold test were $\xi_{2\,(\text{in situ})} = 10.07$ in geographic coordinates and $\xi_{2(\text{tilt corrected})} = 1.494$ after tilt correction, whereas the critical value was $\xi_c = 6.155$ at the 95 per cent confidence level. In the fold test of Enkin (2003), optimal clustering of palaeomagnetic directions occurred at 102 ± 13 per cent unfolding. The mean direction of 28 sites was calculated based on the location of Dulan County (36.4° N, 98.2° E).

319.7° E, $A_{95} = 5.4°$; 50.1° N, 197.2° E, $A_{95} = 7.3°$, Chen *et al.* 2002; Sun *et al.* 2006). Thus, we consider that this direction represents a primary magnetization component acquired during the Middle Jurassic.

6. DISCUSSION

Our palaeomagnetic results provide the first Middle Triassic and Middle Jurassic palaeopoles from the Qaidam Terrane. We conducted a preliminary APWP based on these data, combined with reported Cretaceous and Carboniferous palaeopoles (Sun *et al.* 2006; Wang *et al.* 2016) (Fig. 9). Although other palaeopoles have been reported from the Qaidam Terrane, including for the Early Carboniferous (Yang *et al.* 1992; Cao *et al.* 2017), Late Carboniferous (Li & McWilliams 1989; Cao *et al.* 2017), Late Permian (Xu *et al.* 2011), Early-Middle Jurassic (Yang *et al.* 1992), Late Jurassic (Yang *et al.* 1992; Halim *et al.* 2003) and Cretaceous (Chen *et al.* 2002), we did not include those palaeopoles in our calculations, for the following reasons. The Carboniferous and Jurassic palaeopoles reported by Li & McWilliams (1989) and Yang *et al.* (1992) are based on very few samples (< 20). The Late Carboniferous palaeopole reported by Cao *et al.* (2017) was subjected to neither fold nor reversal tests due to the monoclinal nature of the bedding in the sampling section, and only reverse polarity data were presented. The Early Carboniferous result of Cao *et al.* (2017) was close to their Late Carboniferous result, and Kmax was achieved at 92.8 per cent unfolding, which suggests that this palaeopole may have recorded remagnetization during the early stage of fold deformation. The Permian result of Xu *et al.* (2011) was obtained from the Tianjun region, which is in the southern Qilian Orogen, and could not be treated as a reference for the Qaidam Terrane. Both the Jurassic result of Halim *et al.* (2003) and the Cretaceous palaeopole of Chen *et al.* (2002) were sampled from regions near the Altyn Tagh fault and have clearly undergone local rotation.

APWPs of the Tarim Block and NCB were then compared with those of the Qaidam Terrane. For the results selection (Table A5, Supporting Information), we used the criteria of Van der Voo (1990), choosing those with a quality factor $Q > 4$ and avoiding local rotation. Furthermore, all poles obtained from detrital sedimentary rocks were corrected for inclination shallowing using the commonly observed *f* value of 0.6, or the Tauxe & Kent (2004) E/I method.

6.1 Relationship between the Qaidam and Tarim blocks

As shown in Fig. 8(a), the Tarim Block was located at a much higher palaeolatitude than the Qaidam Terrane, and its northward

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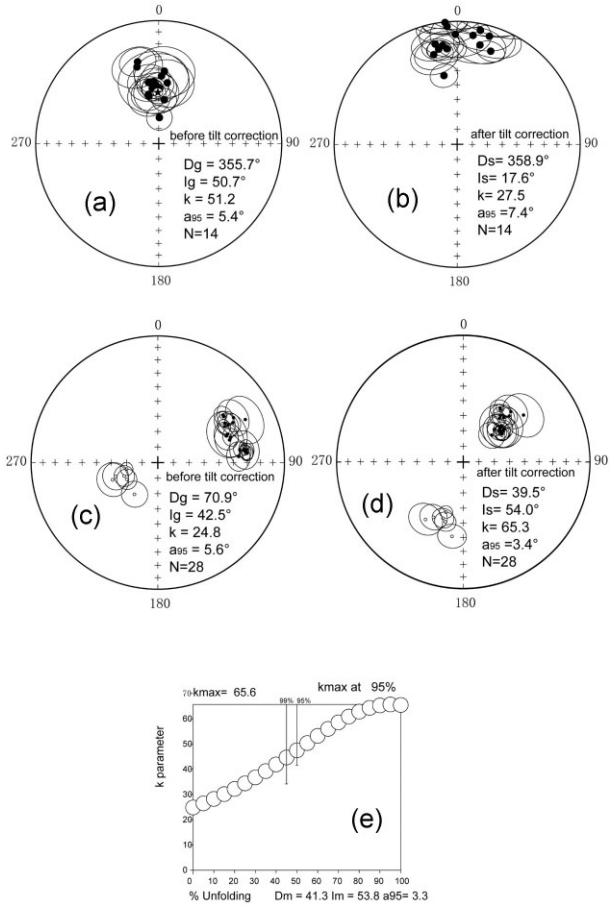


Figure 6. (a) and (b) Equal-area projections of site-mean directions from the Middle Triassic samples of the Qaidam terrane, before and after tilt correction for the low-temperature/low-coercivity, and (c) and (d) high-temperature/high-coercivity components, and before and after tilt correction for the high unblocking temperature component. D_g/I_g, declination/inclination in geographic coordinates; D_s/I_s, declination/inclination in stratigraphic coordinates; k, precision parameter and α_{95}, half-angle of the cone of 95 per cent confidence. The star in (a) is the present geomagnetic field direction.

motion during the Carboniferous was followed by an interval of stable palaeolatitude from the Late Carboniferous to the Cretaceous. During this interval, both the palaeolatitudes and APWP were similar to those of Eurasia, as proposed by Torsvik *et al.*(2012). As mentioned by numerous authors (Zhao *et al.* 1990; Bazhenov *et al.* 1993; Gilder *et al.* 2008; Wu *et al.* 2017; Huang *et al.* 2018), the amalgamation of the Tarim-Junggar Block and the stable Eurasian continent occurred during the Permian to Middle Triassic, which was followed by a long interval of stable palaeolatitude. However, the Qaidam Terrane experienced large-scale northward motion between the Early Carboniferous and Middle Triassic, and it reached a palaeolatitude similar to that of Tarim-Eurasia during the Triassic, which was followed by an interval of stable palaeolatitude that continued until the Cretaceous. This suggests that the Qaidam and Tarim blocks remained separate during their northward drifting, until the Middle Triassic at the latest.

There is a lack of Permian and Triassic sedimentary deposits and palaeomagnetic results for the Qaidam Terrane and thus it is difficult to determine the exact timing of its collision with the Tarim Block. Fortunately, this collision time can be assigned to the Middle–Late Permian based on the angular unconformity in the southern margin of the Tarim Block and syncollisional granites in the northwestern margin of the Qaidam Terrane. Based on drilling and seismic data from the Yubei-Tangbei area in the southern Tarim Block (Zhang et al., 2021), the Carboniferous and Permian strata were deformed to an equally thick anticline structure, while the overlying Triassic strata experienced less structural deformation. On the other side, granitic intrusive rocks developed in the northern margin of the Qaidam Terrane, with ages of 260–275Ma (Gehrels *et al.* 2003a; Wu *et al.* 2009). These I-type granites are metaluminous to peraluminous and form a calc-alkaline series, and they have mixed mantle and crustal sources. Wu *et al.*(2009) suggested that these granitoids are transitional between island arc granites and adakites, indicating their formation within a subduction zone.

6.2 Relationship between the Qaidam Terrane and the NCB

Reference to Fig. 8(b) shows that the NCB underwent large-scale northward motion, with minor clockwise rotation, from the Carboniferous through Middle Triassic, followed by westward motion after the Late Triassic, as noted by Huang *et al.* (2008) and Yang *et al.* (1992). Subsequently, from the Late Triassic through Early Cretaceous, it underwent substantial counterclockwise rotation, whereas the Qaidam Terrane experienced minor clockwise rotation. This difference suggests that the Qaidam Terrane and NCB did not belong to a united dynamic tectonic unit from the Early Carboniferous through Middle Triassic. The overlapping Middle Jurassic palaeopoles of these two tectonic units suggest that the completion of the amalgamation process and the initial connection likely occurred between the Middle Triassic and Middle Jurassic.

However, it is still difficult to propose a detailed convergence process between the NCB and Qaidam Terrane. Because of the discordancy between the palaeomagnetic results for NCB and Tarim-Eurasia revealed by Enkin *et al.* (1992), and a megashear between the NCB and the western terranes/blocks was proposed to reconcile these differences, although no geological evidence was presented. Our results indicate a close relationship between the Qaidam and Tarim blocks since the early Mesozoic. Thus, the region linking these three blocks—the Alxa Terrane—is important for reconciling these tectonic processes. However, the controversy regarding the affiliation between the Alxa Terrane and the NCB cannot be ignored in any discussion of the tectonic evolution of the Alxa Terrane. This terrane was thought to be part of the NCB since the Ordovician (Huang *et al.* 1999), based on consistent palaeopoles. However, Yuan & Yang (2015b) argued that the Alxa Terrane experienced ∼20° counterclockwise rotation and migrated to the NCB after the Early-Middle Triassic, based on the comparison of APWPs from these two tectonic units. Unfortunately, most of the Alxa Terrane is occupied by desert, and its southern (Northern Qilian) and eastern margins (the Bayanwula tectonic belt) experienced intensive compression thrusting and structural shortening during the Cenozoic (Darby & Ritts 2002; Yang *et al.* 2007; Zhang *et al.* 2013b, 2022). No direct geological evidence has been found to reveal the details of the rotation process. Considering the significantly different APWPs of these two tectonic units, we adopted the tectonic relationship proposed by Yuan & Yang (2015b). These two tectonic units should be geographically linked and underwent relative rotation in the Triassic to reach a unified dynamic unit.

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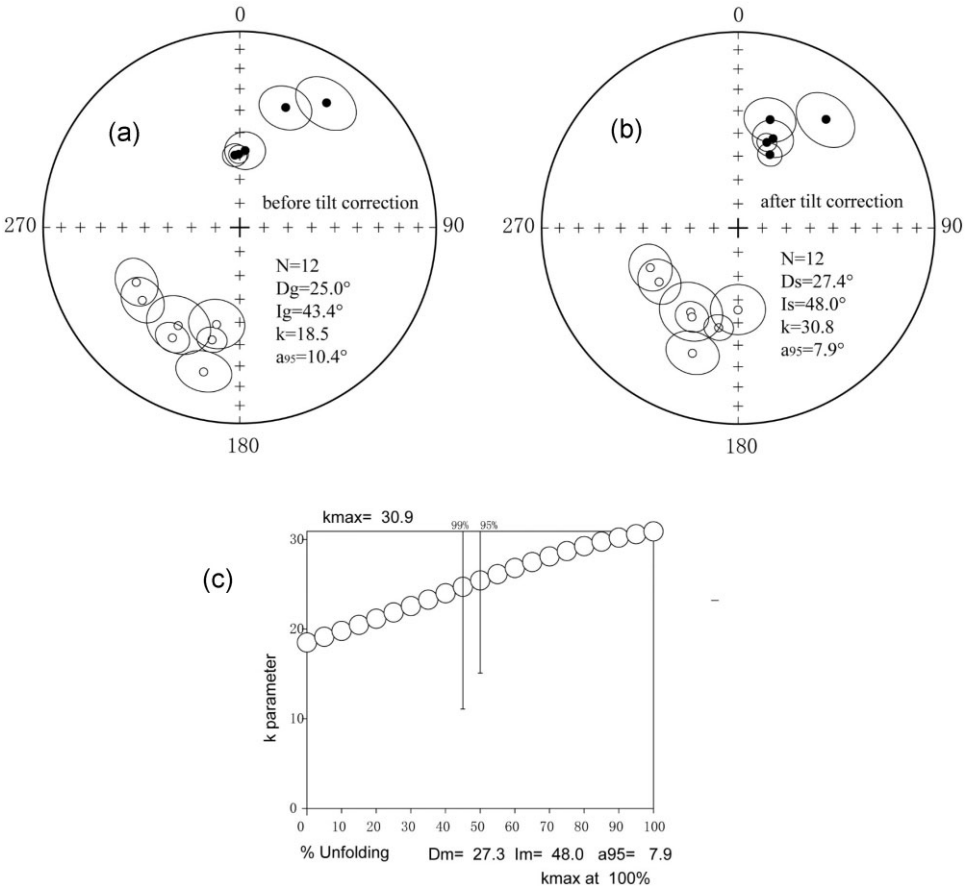


Figure 7. (a) and (b) Equal-area projections of site-mean directions from the Middle Jurassic samples of the Qaidam terrane, before and after tilt correction. D_g/I_g, declination/inclination in geographic coordinates; D_s/I_s, declination/inclination in stratigraphic coordinates; k, precision parameter and α_{95}, half-angle of the cone of 95 per cent confidence.

Table 2. ChRM directions for Middle Jurassic strata in the Qaidam terrane.

Site	Lat. (° N)	Long. (° E)	Strike/dip	*N*/*n*	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)
HJ1	37°00.485′	96°26.283′	133/10	8/10	242.0	−40.1	245.6	−49.4	36.4	9.3
HJ2	37°00.485′	96°26.283′	127/7	11/12	193.8	−40.6	191.0	−47.0	66.2	5.7
HJ3	37°00.485′	96°26.283′	132/12	4/10	233.1	−38.0	235.6	−49.7	102.0	9.1
HJ4	37°00.483′	96°26.211′	138/9	7/10	212.0	−40.6	209.3	−49.2	24.1	12.6
HJ5	37°00.483′	96°26.211′	138/14	6/10	211.3	−34.4	207.4	−47.6	105.2	6.6
HJ15	37°01.632′	96°26.858′	359/17	6/10	356.1	59.7	23.4	56.5	189.3	4.9
HJ16	37°01.632′	96°26.858′	340/15	7/11	359.1	59.4	18.5	52.0	246.4	3.9
HJ17	37°01.632′	96°26.858′	340/15	7/11	4.0	57.7	21.3	49.5	56.2	8.1
HJ21	37°00.584′	96°26.190′	52/13	3/10	194.0	−25.0	200.0	−32.5	156.4	9.9
HJ24	37°00.584′	96°26.190′	150/13	8/11	193.4	−47.8	180.1	−55.6	26.6	10.9
HJ28	37°00.311′	96°25.954′	73/10	7/10	34.8	23.6	38.9	29.5	28.6	11.5
HJ30	37°00.311′	96°25.954′	147/9	7/12	20.9	34.8	16.4	41.9	38.0	9.9
Mean				12	25.0	43.4	27.4	48.0	30.8	7.9

Notes. Lat/Long: latitude and longitude of the sampling site; *N*/*n*: number of samples used to calculate the mean/measured palaeomagnetic direction; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigaphic coordinates, respectively; k: precision parameter and α_{95}: half-angle of the cone of 95 per cent confidence about the mean direction after tilt correction. The site mean direction for this single component passed fold test (Enkin 2003), and optimal clustering of palaeomagnetic directions occurred at 116 ± 49 per cent unfolding. The result of the reversal tests (McFadden & McElhinny 1990) was positive. The mean direction of normal direction is *Dm*/*Im* = 24.5/46.2, $n = 5$ and $k = 42.1$, while the mean direction of reversed direction is *Dm*/*Im* = 209.6/−49.2, $n = 7$ and $k = 23.7$, the angle between the mean directions γ =4.6° <.γ critic = 16.8° indicates a C class result.

On the other hand, geological evidence from the western margin of the Alax-NCB, could provide constraints on the collision of the Qaidam Terrane, as follows. (1) The Permian and Lower–Middle Triassic alternating marine and continental strata in the southern Qilian Shan were in conformable contact with each other, and disconformable contact with the Upper Triassic continental strata (BGMRQP 1991). (2) A narrow ductile sinistral strike-slip shear zone (the Zongwulong tectonic belt) preserved between the

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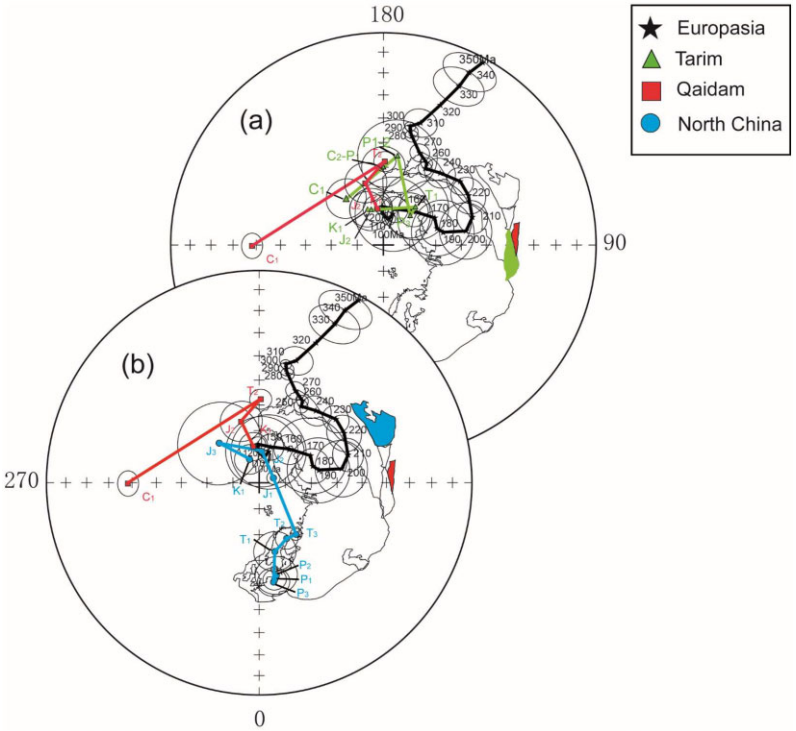


Figure 8. Comparison between the APWP of the Qaidam terrane, Tarim Block, NCB and Eurasia. For clarity, these APWPs are divided into two groups and compared separately.

Qaidam Terrane and the Qilian tectonic belt, was shown to have formed during 250.5–243.3 Ma and was reactivated during 169.7–160.6 Ma, based on the $^{40}Ar/^{39}Ar$ plateau age of K-feldspar and biotite in mylonite (Xu *et al.* 2002). (3) Several granitic intrusives with ages of 249–238Ma occur within the Zongwulong tectonic belt (Guo *et al.* 2009; Menold, 2006; Peng *et al.* 2016). These weak peraluminous I-type granites of middle-high potassium calc-alkaline series, had mixed mantle and crustal sources, and were produced by the collision between the Qaidam Terrane and the Alxa-NCB at around the Middle Triassic.

Based on the timing of the collisions between the Qaidam Terrane, Tarim and NCB, we present a formation process for the North China tectonic domain, based on a synthesis of the APWPs of relevant blocks and published geological evidence. This process can be divided into three phases: first, the formation of the northern boundary. From the Carboniferous through Triassic, the NCB, and Alxa and Tarim blocks had sutured on the southern margin of the CAOB, forming the Solonker Suture, Enger Us ophiolitic melange belt, and the Tianshan suture zone (Xiao *et al.* 2003, 2013, 2015; Han et al. 2010; Gong *et al.* 2018). This process was dated to 332–325 Ma from ophiolites in the northern Tianshan Suture Zone (Xu *et al.* 2006a,b); to 233–226 Ma from eclogites (Zhang *et al.* 2007); to the Middle Permian from radiolarian fossils in the SW Tianshan (Li et al. 2002); and to ∼262–328 Ma, based on U-Pb dating of the gabbro and ultramafic-mafic rock belt in the northern Alxa Block (Gong *et al.* 2018). These results suggest that the Tarim-Alxa-NCB Terrane/block collage had already formed part of the CAOB and the northern boundary of the Palaeo-Tethys Ocean. During this period, the late Carboniferous ophiolites with Rb/Sr ages of 318–331 Ma (Wang *et al.* 2001) and deep water flysch (BGMRQP 1991) from the southern margin of the Qilian Orogenic belt indicate that the Qaidam Terrane was separated from the Qilian-Alxa tectonic unit by a narrow or limited ocean basin during the middle Carboniferous. Second, the Qaidam Terrane was connected to the southern margin of the Tarim Block, producing syncollisional granites during the middle–late Permian (Fig. 9a), as mentioned above. Third, the Qaidam Terrane was forced to slip eastward along the Altyn Tagh Fault, due to the pressure of the subduction of the Palaeo-Tethys branch (Kunlun Ocean) and was connected to the Qilian–Alxa–NCB around the middle Triassic. During this phase, the subduction of the Kunlun Ocean produced subduction-related volcanic rocks in the Kunlun Orogenic belt and on the southern margin of the Qaidam Terrane with ages of 243–236 Ma (Chen *et al.* 2010, 2015; Wu *et al.* 2010; Wang *et al.* 2015; Dong *et al.* 2018). On the northern margin, sinistral strike-slip movement and time of activity of the Altyn Tagh Fault were confirmed by the orientation of syntectonic growth or the syntectonic resetting of minerals in the Altyn Tagh Fault, which produced $^{40}Ar/^{39}Ar$ ages of 223–244 Ma (Li *et al.* 2001; Liu *et al.* 2007). On the eastern margin, a ductile sinistral strike-slip shear zone, syn-collisional granites and a disconformable contact were produced(Fig. 9b). Subsequently, the Qaidam Terrane formed part of the Northern China Domain, and it underwent multiple tectonic uplift in response to collisions with the Qiangtang Terrane, the Lhasa Terrane, and the India Plate. Based on the fission track dating of zircons and apatites, multiple cooling events (221 ± 22 to 96 ± 4 Ma, 40 ± 10 Ma, late Oligocene–Miocene, 9–5 Ma) of the slowly cooling Qaidam Terrane have been revealed (Jolivet *et al.* 2001), which are consistent with multiple uplift events of the Qilian Orogenic belt (Late Triassic–Early Jurassic, Late Jurassic–Early Cretaceous, and Cenozoic) in the Qilian Shan (Pen *et al.* 2018; Lin *et al.* 2019), and with the strike-slip movement and eastern expansion of the Altyn Tagh Fault (232–226, 164.3–178.4, 85–100, 25–40 and 8–10 Ma, Li *et al.* 2001; Liu *et al.* 2007). During this process, intense tectonic deformation occurred mainly on the margins of the

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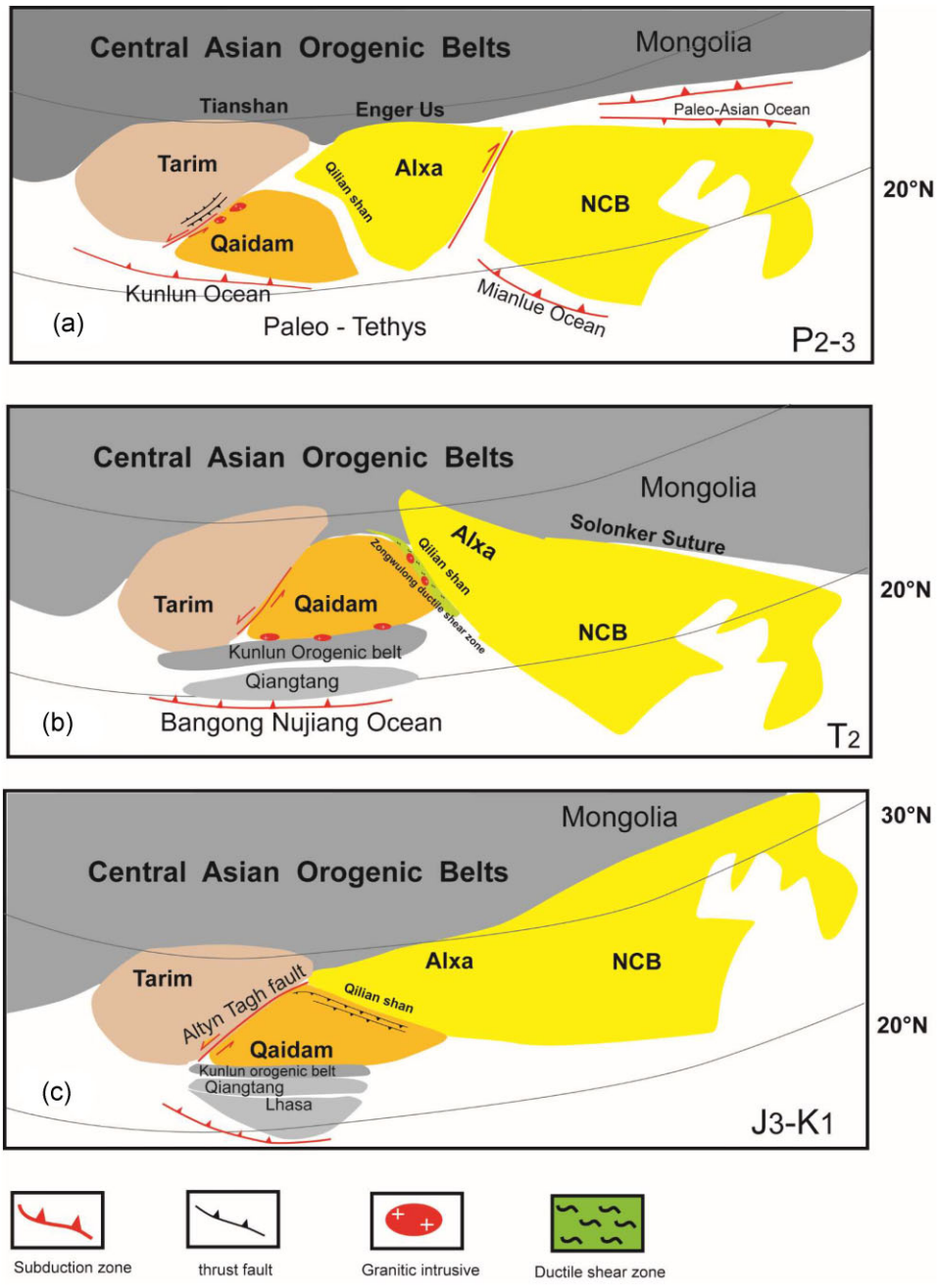


Figure 9. Schematic of the tectonic evolution of the Northern China tectonic domain.

Qaidam Terrane, characterized by arcuate structures near the Altyn Tagh Fault (Dupont *et al.* 2004) and by imbricated thrust faults and related folds (Zhang *et al.* 2020) on the northern edge (Fig. c).

7. CONCLUSIONS

New palaeomagnetic directions were obtained from Middle Triassic lava sites and Middle Jurassic sedimentary strata in the Qaidam Terrane, which passed the fold and reversal tests and were confirmed as the ChRM. These new results corresponded to a Middle Triassic pole (57.6° N, 178.2° E, $A_{95} = 4.0°$) and Middle Jurassic pole (65.8° N, 197.6° E, $A_{95} = 7.8°$). Combined with other reliable data, the APWP of the Qaidam Terrane was established and compared to those of the Tarim and NCB blocks. A synthesis of the motion characteristics indicated by the APWP with other reported geological evidence indicated that the Qaidam Terrane was an independent dynamic unit, distinct from the NCB and Tarim Block, during the late Palaeozoic, until its connection with the Tarim Block around the late Permian. Subsequently, the Qaidam Terrane underwent eastward motion along the boundary between the Tarim Block and Qaidam Terrane, under the influence of the subduction of the Palaeo-Tethys Ocean, and was connected to the NCB in the Middle Triassic, which resulted in the initial formation of the Northern China Domain. After the continued eastward motion of the Qaidam Terrane, with multiple uplifts and tectonic shortening during the Mesozoic and Cenozoic, and multiple collisions with the Qiangtang Terrane, Lhasa Terrane and the India Plate, its modern tectonic configuration was achieved.

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SUPPORTING INFORMATION

Supplementary data are available at *GJI* online.

Table A1. SHRIMP U-Pb zircon data for sample DLT01.
Table A2. SHRIMP U-Pb zircon data for sample WD01.
Table A3. SHRIMP U-Pb zircon data for the sample WD02.
Table A4. LA-ICP-MS zircon data for the DLTB02 sample.
Table A5. The Late Palaeozoic and Mesozoic palaeomagnetic data from the North China, Tarim and Qaidam blocks.
Table A6. Paleomagnetic data from Triassic and Jurassic strata in the Qaidam Basin

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DATA AVAILABILITY

Data to support this paper are uploaded as supporting tables (Tables A1–A6), which can be viewed online from the public repository (https://figshare.com/articles/dataset/Geochronological_and_paleomagnetic_data_from_Triassic_lava_in_the_Qaidam_Basin/11871576).

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